

# Application Reference Handbook

Complete guide to every page, field, button, setting, and calculation

*Field Service / Jobs Management Platform*

Generated from the live codebase • Internal reference

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# 1. Overview & Tech Stack

This application is a field-service / jobs management platform used to schedule jobs, track technicians and marketers, calculate pay splits and office profit, manage client and company records, run balance reports, and send automated messages.

## Technology stack

- **Frontend:** React 19 + TanStack Start (file-based routing, SSR) built with Vite.
- **Styling:** Tailwind CSS with semantic design tokens and shadcn/ui components.
- **Backend:** Lovable Cloud (Postgres database, auth, storage, edge functions).
- **Server logic:** TanStack server functions + public API routes for webhooks/cron.
- **AI:** Gemini-2.5-flash for parsing job messages.
- **Integrations:** Twilio SMS, email queue, geocoding/maps.

## Main navigation

- Dashboard (home) — time-aware greeting + quick overview.
- Jobs · Schedule · DataBoard · Balances · Reports.
- Clients · Companies · Technicians · Installers.
- Settings (admin / manager) and user profile.

## Security model

Roles are stored in a dedicated `user_roles` table (never on the profile) and checked via a security-definer `has_role()` function. Row-Level Security governs every table; the Data API requires explicit GRANTS per table.

## 2. Core Calculations & Formulas

All money metrics only count when a job's status is **Completed**. Scheduled, pending, and cancelled jobs contribute \$0 to revenue, profit, tech pay, and marketer pay. The collected deposit is stored separately and is never part of price, so it is never double-counted.

### Revenue (net amount that funds the splits)

```
revenue = price
         - co_parts
         - office_parts
         - parts
         - tip
         - (cc_fee if payment is card / credit card)
```

### Pay splits

Each party (marketer, tech, office) can be paid by **percent** of revenue or a **fixed** amount.

```
marketerPct = marketer_pct / 100
techPct     = tech_pct / 100

marketerBase = (mode == fixed) ? marketer_fixed_amount : revenue * marketerPct
techBase     = (mode == fixed) ? tech_fixed_amount     : revenue * techPct
officeBase   = (mode == fixed) ? office_fixed_amount   : revenue * max(0, 1 - marketerPct - techPct)
```

### Stored totals (rounded to cents)

```
total_marketer = marketerBase + co_parts
total_tech     = techBase + parts + tip
total_office   = officeBase + office_parts
```

Parts the marketer fronted (co\_parts) are reimbursed via the marketer total; tech parts and the customer tip flow to the tech; office parts flow to the office.

### Marketer balance (office perspective)

Positive net = the office owes the marketer; negative = the marketer owes the office.

```
if marketer_collected: net = total_marketer - price
else:                  net = total_marketer

# Parts charges (flat fees for parts bought for a company)
# reduce what the office owes the marketer:
net = net - sum(parts_charges for that marketer in range)
```

- **Default** (office/tech collected payment): office owes the marketer their earned share.
- **marketer\_collected = true**: the marketer holds the whole price, so they owe back everything except their share.
- **Parts charges**: companies owe the office for parts purchased on their behalf, lowering the net.

## DataBoard metric values

| Metric              | Formula (completed jobs only)              |
|---------------------|--------------------------------------------|
| Revenue             | price                                      |
| Office profit       | total_office (already net of parts/cc fee) |
| Tech pay            | total_tech                                 |
| Marketer pay        | total_marketer                             |
| Parts cost          | parts + office_parts + co_parts            |
| Tip / Cost / CC fee | tip / cost / cc_fee                        |
| Count               | always 1 per job                           |

## 3. Jobs — Fields, Table & Actions

### Add / Edit Job form

Core fields captured when creating or editing a job (plus admin-defined custom fields stored in `extra_fields`):

| Field                           | Purpose                                                        |
|---------------------------------|----------------------------------------------------------------|
| Client / Phone / Address        | Customer identity and location (address validated & geocoded). |
| Company / Company_1             | Marketer / company name used for balance grouping.             |
| Job type / Status               | Categorization and workflow state.                             |
| Job date / time / end           | Scheduling window.                                             |
| Tech name / Installer           | Assigned technician and installer.                             |
| PO number / Maps                | Reference number and map link.                                 |
| Price                           | Customer charge (drives revenue).                              |
| co_parts / office_parts / parts | Parts fronted by marketer / office / tech.                     |
| Tip / CC fee / Cost / Payment   | Add-ons, card fee, internal cost, payment method.              |
| Pay modes & amounts             | percent vs fixed for marketer / tech / office.                 |
| manual_percentage               | Manual tech % override.                                        |
| marketer_collected              | Marks the marketer collected the full price.                   |
| Deposit fields                  | amount, date, method, check #, received flag.                  |
| Notify settings                 | channels, enabled, lead minutes list.                          |
| Notes                           | Free-text job notes.                                           |

### Jobs table

- Sortable, filterable columns with show/hide column toggle.
- Inline editing of cells (EditableCell) writes directly to the job.
- Status badges with colors from the configurable status list.
- Bulk edit bar for multi-row updates and bulk actions.
- Filters: date range, marketer, status, tech, search.
- Per-row actions: edit, send message, parse message, reschedule.

## 4. DataBoard

A customizable analytics board of draggable widgets backed by the shared metrics module, so values always match what the rest of the app stores.

### Widget types

- KPI — single headline metric.
- Chart — trends and breakdowns.
- Table — tabular slices of jobs.
- Insight — auto-generated highlights (configurable).
- Goal — progress toward a target.
- Calendar / Activity / Map — schedule, recent activity, and geographic view.

### Controls

- Add Widget menu, View Templates, Time Range bar, Filters bar.
- Drag to rearrange widgets on desktop.
- **Mobile drag-lock:** widgets are locked from dragging on touch devices so scrolling never moves them.
- Export board dialog for sharing the current view.

## 5. Schedule & Reminders

- Calendar view of jobs by date/time.
- Reschedule dialog to move a job to a new date/time.
- Job reminders dispatched via a public cron hook (dispatch-job-reminders).
- Per-job notify settings: channels, enabled flag, and lead-minute list.



## 6. Clients & Companies

### Clients

- Fields: name, phone, email, address, notes.
- Import Clients dialog for bulk import.

### Companies (marketers)

- Fields: company\_name, contact\_name, email, percentage, company\_type[] (multi).
- Marketer type select drives how splits default.
- Percentage feeds the default marketer split.

## 7. Technicians & Installers

### Technicians

- Fields: tech\_name, percentage, phone\_number, city, pincode, linked user\_id.
- Percentage feeds the default tech split.

### Installers

- Fields: name, email, phone\_number, install\_types[] (multi).

### Installation catalog & editor

- Catalog: groups, models (with colors), sizes, sub-items, colors.
- Job Installations editor attaches system type, model, size, color, parts PO, notes, and sub-items to a job.

## 8. Balances & Parts Charges

The Balances page groups completed jobs by marketer over a date range and computes the net amount owed (see Section 2 for the formula).

### Balance table columns

- Marketer · Jobs count · Earned (total\_marketer) · Parts charged · Net · Status.
- Status label: 'Office owes marketer', 'Marketer owes office', or 'Settled'.
- Per-marketer PDF statement export.

### Parts Charges card

Flat fees for parts the office buys on a company's behalf — these are **not** jobs. Each charge has marketer, amount, charge\_date, and an optional description. Charges reduce the marketer's net (the company owes the office).

- Add / edit / delete charges via dialog.
- Included in balance net and in the per-marketer PDF statement.

## 9. Reports & Automations

### Report builder

Builds a configurable PDF (and HTML email) from a shared report model.

#### Sections (ordered, each toggleable)

| Section         | Contents                                           |
|-----------------|----------------------------------------------------|
| Title           | Report title.                                      |
| Date Range      | Resolved range text + total job count.             |
| Totals          | Revenue, Tech, Office, Marketer (each toggleable). |
| Balance summary | Per-marketer net table + grand net.                |
| Parts charges   | Charges table (company owes office) + total.       |
| Jobs Table      | Selected columns for filtered jobs.                |

#### Other settings

- Column picker (date, client, marketer, tech, PO, type, status, price, parts, totals, payment, paid, etc.).
- Filters: marketers, statuses, and date mode.
- Date modes: all, custom, today, this/last week, this/last month, this year (resolved at render time).

### Automations

Scheduled reports stored in `report_automations`.

- Schedule: daily / weekly (weekday) / monthly (month day) + time + timezone.
- Recipients: roles, marketers, emails, per-marketer split, and send-to-marketer option.
- Dispatched by the public cron hook (`dispatch-report-automations`), which renders HTML email.

## 10. Messaging & AI

### Templates & sending

- Message Templates manager with reusable templates.
- Template variables (30+) such as client, address, job date/time, tech, price, PO, status.
- Send Message dialog (SMS via Twilio); email queue for outbound email.

### AI message parsing

- Parse Message dialog reads a pasted message and extracts job fields via Gemini-2.5-flash.
- Diff preview shows proposed changes before applying.
- Match review reconciles parsed data with existing jobs/clients.
- Applied updates recompute totals using the shared job-calc logic.

## 11. Settings

- **Job Form Builder** — define custom fields stored in extra\_fields.
- **Statuses manager** — status names, colors, ordering.
- **Type groups manager** — job-type groupings.
- **Door centers manager** — pickup locations.
- **Installation catalog** — groups, models, colors, sizes, sub-items.
- **Message templates manager** — templates and variables.
- **Users manager** — invite users, assign roles, manage permissions.
- **My profile** — name, contact, timezone, avatar.

## 12. Database Schema

Every table and column in the public schema. Boolean/number/text/json/uuid noted per column.

### jobs

| Column                    | Type      | Column                   | Type      |
|---------------------------|-----------|--------------------------|-----------|
| id                        | uuid      | co_parts                 | number    |
| created_at                | timestamp | office_parts             | number    |
| updated_at                | timestamp | parts                    | number    |
| created_by                | uuid      | tip                      | number    |
| client_id                 | uuid      | payment                  | text      |
| company_id                | uuid      | paid                     | boolean   |
| installer_id              | uuid      | manual_percentage        | number    |
| pickup_door_center_id     | uuid      | marketer_pay_mode        | text      |
| address                   | text      | marketer_fixed_amount    | number    |
| maps                      | text      | tech_pay_mode            | text      |
| phone_no                  | text      | tech_fixed_amount        | number    |
| company                   | text      | office_pay_mode          | text      |
| company_1                 | text      | office_fixed_amount      | number    |
| comp_type                 | text      | marketer_collected       | boolean   |
| job_type                  | text      | total_marketer           | number    |
| status                    | text      | total_tech               | number    |
| job_date                  | date      | total_office             | number    |
| job_time                  | text      | deposit_amount           | number    |
| job_time_end              | text      | deposit_date             | date      |
| scheduled_completion_date | date      | deposit_payment_method   | text      |
| completed_at_date         | date      | deposit_check_no         | text      |
| tech_name                 | text      | deposit_received         | boolean   |
| installer_name            | text      | notify_enabled           | boolean   |
| po_number                 | text      | notify_channels          | text[]    |
| check_no                  | text      | notify_lead_minutes      | number    |
| notes                     | text      | notify_lead_minutes_list | number[]  |
| price                     | number    | notified_lead_minutes    | number[]  |
| cost                      | number    | notified_at              | timestamp |
| cc_fee                    | number    | extra_fields             | json      |

## parts\_charges

| Column      | Type      |
|-------------|-----------|
| id          | uuid      |
| marketer    | text      |
| amount      | number    |
| charge_date | date      |
| description | text      |
| created_at  | timestamp |
| updated_at  | timestamp |

## clients

| Column     | Type      |
|------------|-----------|
| id         | uuid      |
| name       | text      |
| phone      | text      |
| email      | text      |
| address    | text      |
| notes      | text      |
| created_by | uuid      |
| created_at | timestamp |
| updated_at | timestamp |

## companies

| Column       | Type      |
|--------------|-----------|
| id           | uuid      |
| company_name | text      |
| contact_name | text      |
| email        | text      |
| percentage   | number    |
| company_type | text[]    |
| created_at   | timestamp |
| updated_at   | timestamp |

## technicians



| Column       | Type      |
|--------------|-----------|
| id           | uuid      |
| tech_name    | text      |
| percentage   | number    |
| phone_number | text      |
| city         | text      |
| pincode      | text      |
| user_id      | uuid      |
| created_at   | timestamp |
| updated_at   | timestamp |

## installers

| Column        | Type      |
|---------------|-----------|
| id            | uuid      |
| name          | text      |
| email         | text      |
| phone_number  | text      |
| install_types | text[]    |
| created_at    | timestamp |
| updated_at    | timestamp |

## job\_installations

| Column                  | Type        |
|-------------------------|-------------|
| id                      | uuid        |
| job_id                  | uuid        |
| system_type             | text        |
| group_id / group_name   | uuid / text |
| model_id / model_name   | uuid / text |
| size_id / size_label    | uuid / text |
| color                   | text        |
| parts_po                | text        |
| notes                   | text        |
| sub_items               | json        |
| sort_order              | number      |
| created_at / updated_at | timestamp   |

## door\_centers

| Column                  | Type      |
|-------------------------|-----------|
| id                      | uuid      |
| name                    | text      |
| contact_name            | text      |
| phone                   | text      |
| address                 | text      |
| notes                   | text      |
| sort_order              | number    |
| created_at / updated_at | timestamp |

## install\_groups / install\_models / install\_sizes / install\_colors / install\_sub\_items

| Column    | Type                                     |
|-----------|------------------------------------------|
| groups    | id, name, sort_order                     |
| models    | id, group_id, name, colors[], sort_order |
| sizes     | id, width, height, label, sort_order     |
| colors    | id, name, sort_order                     |
| sub_items | id, group_id, name, sort_order           |

## profiles

| Column                                | Type      |
|---------------------------------------|-----------|
| id                                    | uuid      |
| display_name / first_name / last_name | text      |
| email / phone / mobile_phone          | text      |
| job_title                             | text      |
| timezone                              | text      |
| avatar_url                            | text      |
| notes                                 | text      |
| created_at / updated_at               | timestamp |

## report\_automations

| Column                  | Type      |
|-------------------------|-----------|
| id                      | uuid      |
| name                    | text      |
| enabled                 | boolean   |
| template                | json      |
| schedule                | json      |
| recipients              | json      |
| last_run_at             | timestamp |
| created_by              | uuid      |
| created_at / updated_at | timestamp |

## app\_settings & user\_preferences

| Column           | Type                                     |
|------------------|------------------------------------------|
| app_settings     | key (text), value (json), updated_at     |
| user_preferences | user_id (uuid), prefs (json), updated_at |

## Auth & roles

| Column           | Type                                                    |
|------------------|---------------------------------------------------------|
| user_roles       | id, user_id, role (enum app_role)                       |
| custom_roles     | name, created_at                                        |
| permissions      | key, label, description                                 |
| role_permissions | role_name, permission_key                               |
| pending_invites  | email, role, token, expires_at, accepted_at, invited_by |
| admin_seed       | email, created_at                                       |

## Email system

| Column                   | Type                                                                        |
|--------------------------|-----------------------------------------------------------------------------|
| email_send_log           | recipient_email, template_name, status, message_id, metadata, error_message |
| email_send_state         | batch_size, send_delay_ms, TTLs, retry_after_until                          |
| email_unsubscribe_tokens | email, token, used_at                                                       |
| suppressed_emails        | email, reason, metadata                                                     |

## 13. Roles & Permissions

Roles live in `user_roles` (enum `app_role`). Permissions are defined in `permissions` and mapped to roles in `role_permissions`, with custom roles supported.

| Capability                          | Typical access                                                                     |
|-------------------------------------|------------------------------------------------------------------------------------|
| View all jobs                       | Admin / Manager (and via <code>has_role</code> / <code>can_view_all_jobs</code> ). |
| Create / edit jobs                  | Authenticated staff.                                                               |
| Edit / delete parts charges         | Admin / Manager.                                                                   |
| Manage users & roles                | Admin.                                                                             |
| Settings (form, statuses, catalogs) | Admin / Manager.                                                                   |
| Run & schedule reports              | Admin / Manager.                                                                   |

Exact per-permission mapping is data-driven via the `permissions` and `role_permissions` tables and can be edited in the Users manager.